

Full Marks: 80
Year: I

Aliah University
Department of Electrical Engineering, B-Tech
Even (Spring) Semester Examination, 2023 (Reg./Supple)
Subject: Basic Electrical Engineering [EENUGES01]

Time: 3 Hours
Semester: II

Answer any five questions

5x16=80 Marks

1. a) State and explain Kirchhoff's laws as applied to D.C circuits. 4 Marks
- b) Derive the necessary expression for energy stored in an Inductance. 4 Marks
- c) For the circuit shown in Fig. 1, determine V_x using nodal analysis. 4 Marks
- d) For the circuit shown in Fig. 2, calculate the current in the $5\ \Omega$ resistance using mesh analysis. 4 Marks

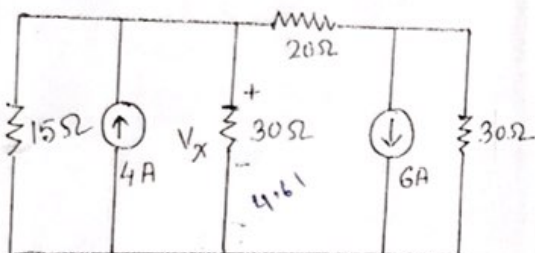


Fig. 1

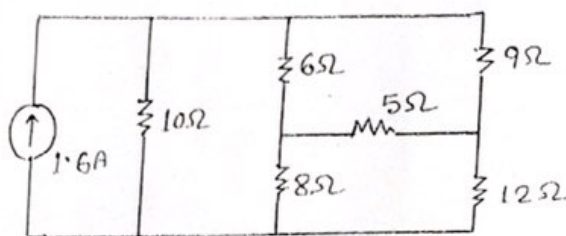


Fig. 2

2. a) The no-load current of a 6000/400 V, single-phase, 50 Hz transformer is 0.05 A. It consumes 80 W power at no-load when rated supply is given to l.v. side and h.v. side is kept open. Calculate the following: (a) Power factor of no-load current (b) Iron-loss component of current (c) Magnetizing component of current. 6 Marks
- b) In a 20 kVA, 2000/200V, 50 Hz single phase transformer, the iron and full load copper losses are found to be 340 W and 410 W, respectively. (a) Calculate the efficiency on u.p.f. at (i) full-load, and (ii) half-load. (b) Determine the Load for maximum efficiency and the iron and copper loss in this case. 6 Marks
- c) Draw the phasor diagram for lagging p.f load in an ideal transformer. 4 Marks
3. a) Derive the expression for rms value of a sinusoidal varying current. 5 Marks
- b) An alternating voltage is given by $V=220\sin 314t$. Calculate i) frequency ii) maximum value iii) average value iv) RMS value. 6 Marks
- c) A capacitor of $20\ \mu\text{F}$ is connected to a voltage supply of 230 V. Find the current flowing in the circuit at frequency (a) 50 Hz and (b) 25 Hz and also give the expression for voltage and current in both cases. 5 Marks
4. a) Write short notes on characteristics of DC shunt generator. 5 Marks
- b) Explain the superposition theorem with a suitable example. 5 Marks
- c) Derive the expression of torque for a DC motor using common notations. 6 Marks
5. a) Draw the connection diagrams of D.C motor for different types. 5 Marks
- b) State the principle of DC generator? What is the need for starter in a DC motor? 2+3 Marks
- c) A DC shunt motor runs at 1200 rpm and takes an input of 700 Watt at 220V under no-load conditions. The shunt-field current is 1A and armature resistance is $0.2\ \Omega$. Find the speed when the machine is used as a generator if the line current is same in both the cases. 6 Marks

6. a) For the circuit shown in Fig. 3 find the Thevenin voltage and Norton current using the principle of superposition. 2x8=16 Marks
- b) Find the Thevenin equivalent of the circuit of Fig. 4 which has one dependent current generator.

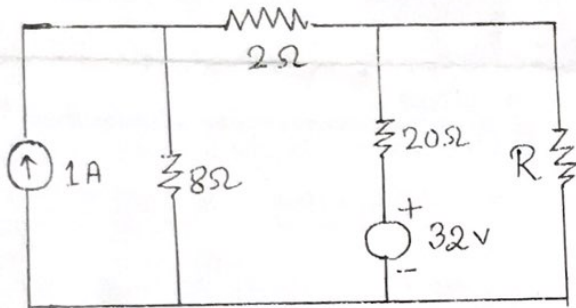


Fig. 3

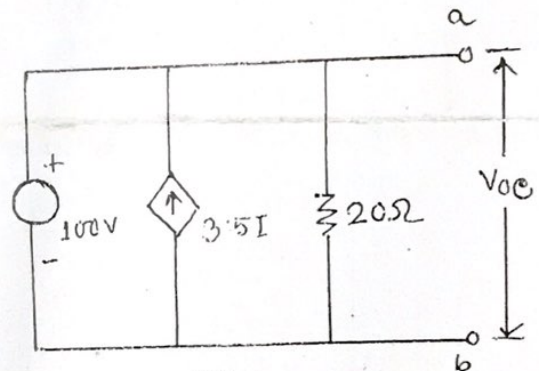


Fig. 4

ALIAH UNIVERSITY
END SEMESTER EXAMINATION, SPRING 2023
PAPER: MATUGBS02 (Engineering Mathematics II)

Full Marks: 80

Time Allowed: 3 Hours

All symbols have their usual meanings

Answer any **EIGHT** questions:

8 × 10 = 80

1. (a) Show that

[5]

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc).$$

(b) Define a row reduced matrix. Give an example of a row reduced matrix.

[5]

2. (a) If a, b, c are all positive real numbers, then prove that

[5]

$$\frac{a^2 + b^2}{a + b} + \frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} \geq a + b + c.$$

(b) Determine the row rank of the matrix

[5]

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 8 & 6 \\ 3 & 6 & 6 & 3 \end{bmatrix}.$$

3. (a) Apply elementary row operations to reduce the following matrix to a row echelon matrix:

[5]

$$\begin{bmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{bmatrix}.$$

(b) Apply row and column operations to find the rank of the matrix

[5]

$$\begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}.$$

4. (a) Solve the system of linear equations by Cramer's rule:

[5]

$$x + 2y - 3z = 1, \quad 2x - y + z = 4, \quad x + 3y = 5.$$

(b) Evaluate $(A^{-1})^2 + A$, where

[5]

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

5. (a) Apply Descartes's rule of sign to examine the nature of the roots of the equation [5]

$$x^4 + 2x^2 + 3x - 1 = 0.$$

- (b) Solve the following equation given two of the roots are equal in magnitude but opposite in sign: [5]

$$2x^3 - x^2 - 18x + 9 = 0.$$

6. (a) In a vector space V over a field F , prove that [5]

$$(i) \ 0\alpha = \theta \text{ for all } \alpha \text{ in } V, \quad (ii) \ c\theta = \theta \text{ for all } c \text{ in } F.$$

- (b) Prove that the set $\{(2, 1, 1), (1, 2, 1), (1, 1, 2)\}$ forms a basis of $\mathbb{R}^3$ over $\mathbb{R}$. [5]

7. (a) State Cayley-Hamilton theorem. Use it to find A^{-1} , where $A = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$. [5]

- (b) Find out the eigenvalues of the matrix [5]

$$\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & -1 \\ 3 & 2 & -2 \end{bmatrix}.$$

8. (a) Find out $|z|$ and $\arg z$ and express it in the polar form, where $z = 1 - i$. [5]

- (b) Evaluate $\int_C z^2 dz$, where C is the straight line joining the origin $O(0,0)$ to the point $P(2,1)$ in the complex plane. [5]

9. (a) Show that the following equation is exact and hence find its solution: [5]

$$(4x^3y^3 - 2xy)dx + (3x^4y^2 - x^2)dy = 0.$$

- (b) Solve: $3e^x \tan y + (1 - e^x) \sec^2 y \, dy = 0$. [5]

10. (a) Solve: $(y^4 - 2x^3y)dx + (x^4 - 2xy^3)dy = 0$. [5]

- (b) Solve: $\frac{dy}{dx} + \frac{4x}{x^2+1}y = \frac{1}{(x^2+1)^3}$. [5]

11. (a) Solve: $p^2 + 2xp - 3x^2 = 0$. [5]

- (b) Find out the general solution and the singular solution (if any) of the following equation: [5]

$$y = px + \sqrt{1 + p^2}.$$

12. (a) Find the general solution of the following differential equation: [5]

$$\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0.$$

- (b) Solve: [5]

$$3\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 8y = 5e^{-3x}.$$

Department of English
End Semester Examination, Spring 2023
ENGUGHU01: Communicative English

Full Marks: 80

Time: 3 Hours

Section I

1. Answer **ANY TEN** of the following questions:

2x10=20 Marks

- ☒ A. What is an Aviary?
- ☒ B. What is the meaning of *Lingua Franca*?
- ☐ C. What do you mean by Intrapersonal Communication?
- ☒ D. What is 'scanning' in reading?
- ☒ E. Define Upward communication?
- ☒ F. What is Grapevine communication?
- ☐ G. What is an Idiom? Give an example.
- ☒ H. What is the full form of LSRW skills?
- ☒ I. What is modal verb? Give an example.
- ☐ J. What is passive listening?
- ☐ K. What is Mass Communication?
- ☒ L. What is feedback in Communication?
- ☐ M. What is Extensive Reading?
- ☒ N. What is Bottom-up Listening Skill?

Section II

2. Answer **ANY FOUR** of the following questions: (word limit 200)

5x4=20 Marks

- ☒ A. What is the difference between Verbal communication and Nonverbal Communication?
- ☒ B. What are the barriers of effective Listening?
- ☐ C. What are the different stages of Reading?
- ☒ D. What are the different strategies of Speaking?
- ☐ E. What are the different levels of Communication?
- ☐ F. What are the demerits of Grapevine communication?

Q. Give examples of Paralinguistic features of voice?

Section III

3. Answer **ANY FOUR** of the following questions: (word limit 400)

10X4=40Marks

- A. Define Communication. What are the different features of Communication?
- B. Attaching a Resume, write a job application for the post of an English Teacher in a reputed School.
- C. Write a report on Blood Donation Camp organised by your department.
- D. As an owner of a reputed sports shop, write a letter to a cricket equipment manufacturing company to order cricket bats and other related equipments required for your shop.
- E. Mention the different presentational strategies of Speaking Skills.
- F. Write an email to your Head of the Department asking him/her to grant you a leave of absence due to a severe illness suffered by you.
- F. Adding a suitable title, write a Precis of the following passage:

It is physically impossible for a well-educated, intellectual, or brave man to make money the chief object of his thoughts just as it is for him to make his dinner the principal object of them. All healthy people like their dinners, but their dinner is not the main object of their lives. So all healthy minded people like making money ought to like it and enjoy the sensation of winning it; it is something better than money.

A good soldier, for instance, mainly wishes to do his fighting well. He is glad of his pay—very properly so and justly grumbles when you keep him ten years without it—till, his main mission of life is to win battles, not to be paid for winning them. So of clergymen. The clergyman's object is essentially baptize and preach not to be paid for preaching. So of doctors. They like fees no doubt—ought to like them; yet if they are brave and well-educated the entire object to their lives is not fees. They on the whole, desire to cure the sick; and if they are good doctors and the choice were fairly to them, would rather cure their patient and lose their fee than kill him and get it. And so with all the other brave and rightly trained men: their work is first, their fee second—very important always; but still second.

ALIAH UNIVERSITY

Name of the Examination: Second SEM

Session: 2022-23

Subject: Engineering Chemistry

Course code: CHMUGBS01

Duration: 3 Hrs

Full Marks: 80

Group A

01. Explain the following terms (any FOUR): (i) Path function (ii) Irreversible process (iii) Molar heat capacity (iv) Enthalpy (v) Extensive property (vi) Homogeneous system. 2.5x4

Answer any **FOUR**

02. (a) What is meant by equilibrium and non-equilibrium states? (b) Write down the criteria of equilibrium. 2.5+2.5
03. (a) What is Joule Thomson phenomenon? (b) Prove that for the Joule Thomson phenomenon is an isenthalpic process. 2.5+2.5
04. (a) Establish the relation between ΔH and ΔU . (b) Calculate the amount of heat necessary to raise 213.5 gm of water from 25° C to 100° C ? 2.5+2.5
05. (a) Derive the first law of thermodynamics and explain each term involved in it. $\Delta E = q + w$ 2.5+2.5
06. (a) Establish the expression of work done in adiabatic reversible expansion of an ideal gas. (b) Prove that work done is more in reversible process than that of irreversible process. 2.5+2.5

Group B

Answer any **TEN**

07. Write short notes on the properties and preparations of Bakelite. 05
08. Give brief account of: (a) Compounding of Rubber (b) Polythene. 2.5+2.5
09. Explain with examples: (a) Conducting polyaniline (b) Gutta Percha. 2.5+2.5
10. (a) What is meant by green chemistry? (b) Write informative notes on the mechanism of radical polymerisation 2.5+2.5
11. (a) Write an essay on various types of synthetic rubber (b) How would you prepare Adipic acid by green method 2.5+2.5
12. (a) Write down the importance of green chemistry. (b) Give the preparation of Aspirin from salicylic acid using microwave technique. 2.5+2.5
13. Write notes on: (a) Polystyrene (PS) (b) Polyvinyl Chloride (PVC). 2.5+2.5
14. (a) What are drawbacks of natural rubber? (b) How will you prepare Novalac resin from phenol and formaldehyde? Explain with mechanism? 2+3
15. (a) What are biopolymers? (b) How is it prepared? (c) Give some importance of biopolymers. 1+2+2
16. (a) Why are silicones called inorganic polymers? (b) Discuss the synthesis procedure of linear chain silicones. 2+3
17. (a) What is meant by degree of polymerisation? (b) Distinguish between addition and condensation polymerization. 2+3
18. How the following are prepared? Mention their uses (a) LDPE (b) Teflon. 2.5+2.5

THE END

Aliah University
End-Semester Examination (Even Semester) - 2023
(CSE IST Year 2ND Semester)

Subject Name: Programming for Problem Solving
Subject Code: CSEUGES01

Full Marks: 80
Time: 3hrs

Group-A
(Answer all questions)

5 × 2 = 10

1.
 - a. Define Tail Recursion.
 - b. What do you mean by `int *a[20]`?
 - c. State the characteristics of an Algorithm
 - d. How two dimensional arrays is declared and initialized?
 - e. What is generic pointer?

Group-B

(Answer any five questions)

5*6= 30

2. Explain logical operators with examples. 6
3. Write short note on `sprintf()`, `putc()`, `getche()`. 6
4. What is function? Discuss call by reference with examples. 2+4=6
5. What is null pointer? Define and declare pointer with examples. 2+4=6
6. Difference between compiler and interpreter. What is role of linker and loader? 3+3=6
7. Consider an array `int marks[20][5]` where base address=1000 and number of words per memory location of the array=2. Compute the address of `marks[10][4]` in row major and column major order. 3+3=6

Group-C

(Answer any four questions)

4 × 10 = 40

8. Define recursive function. Write a program using recursion to calculate Fibonacci series 2+8=10
9. What is nested Structure? Write a program to read and display the information (name, emp no, address and ph no.) of 10 employees. 2+8=10
10. State importance of header file. Explain different types of loops with example in C. 2+8=10
11. What is File? Write a program to Create a File, Write in it and Close the File. 2+8=10
12. Differentiate between array and structure. Define Union and how members of a union are accessed using a program code 3+2+5=10

$$3+(4) + 8 + 2 + 6 + 2 + 6 + 6 = 37.$$